



DG INTERNS HUB — CYBERSECURITY INTERNSHIP

WEEK 2 · TASK 6

Vulnerable Lab Setup Report

Metasploitable 2 & DVWA — Target Environment Setup

INTERN	Mah Noor
PROGRAM	TechBiz Academy – Cybersecurity Track
Intern Code	TBS-2026-242
TASK WINDOW	Aug 29, 2026, 5:00 AM – Aug 31, 2026, 10:00 PM (PKT)
REPORT DATE	August 31, 2026

1. Objective

The objective of this task was to build a safe, isolated practice environment for offensive security training by deploying an intentionally vulnerable target machine, connecting it to an attacker machine (Kali Linux) on the same internal network, and verifying network-layer connectivity between the two. In addition, the free TryHackMe DVWA room was completed to gain browser-based, guided practice against the Damn Vulnerable Web Application (DVWA).

2. Tools & Environment Used

Hypervisor	VMware Workstation / Player
Attacker Machine	Kali Linux (VMware VM)
Target Machine	Metasploitable 2 (intentionally vulnerable Linux VM)
Online Practice Room	TryHackMe — DVWA (free room, browser-based AttackBox)
Network Mode	Host-only / NAT (both VMs on the same internal subnet)

3. Steps Performed

3.1 Downloading and Importing Metasploitable 2

- Downloaded the Metasploitable 2 disk image and extracted it to a local folder.
- Opened VMware and used File → Open to import the Metasploitable virtual machine configuration file (.vmx).
- The VM was added to the VMware library and powered on.

3.2 Configuring the Network

- Both the Kali VM and the Metasploitable VM network adapters were set to the same network mode (Host-only / NAT) in VM Settings, ensuring the two machines could reach each other on an internal, isolated subnet rather than the public internet.

3.3 Logging in and Identifying the Target IP

Metasploitable 2 was booted and logged into using the default credentials:

Username	msfadmin
Password	msfadmin

Once logged in, the command "ifconfig" was run to identify the machine's IP address on the internal network.

```

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:50:b2:92
          inet addr:192.168.139.129 Bcast:192.168.139.255 Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe50:b292/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:53 errors:0 dropped:0 overruns:0 frame:0
          TX packets:67 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5041 (4.9 KB)  TX bytes:7154 (6.9 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:113 errors:0 dropped:0 overruns:0 frame:0
          TX packets:113 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:29705 (29.0 KB)  TX bytes:29705 (29.0 KB)

msfadmin@metasploitable:~$

```

Figure 1 — Metasploitable 2: "ifconfig" output showing target IP address 192.168.139.129

3.4 Identifying the Kali (Attacker) IP

On the Kali Linux VM, the command "ip a" was run to confirm Kali's own IP address and verify that it was on the same subnet as the Metasploitable target.

```

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.139.128 netmask 255.255.255.0 broadcast 192.168.139.255
    inet6 fe80::20c:29ff:fe58:3ff6 prefixlen 64 scopeid 0<link>
    ether 00:0c:29:58:3f:f6 txqueuelen 1000 (Ethernet)
    RX packets 1563037 bytes 2296990678 (2.1 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 284201 bytes 18701538 (17.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 30943 bytes 24720414 (23.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 30943 bytes 24720414 (23.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(mahnoor@kali) - [~/Desktop/techbiz]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:58:3f:f6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.139.128/24 brd 192.168.139.255 scope global dynamic noprefixroute eth0
        valid_lft 1752sec preferred_lft 1752sec
    inet6 fe80::20c:29ff:fe58:3ff6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 8e:16:de:d3:23:6d brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::8c16:deff:fed3:236d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

Figure 2 — Kali Linux: "ip a" output showing attacker IP address 192.168.139.128, on the same subnet as the target

3.5 Verifying Connectivity (Ping Test)

From the Kali terminal, the target was pinged using:

ping 192.168.139.129

The ping ran successfully and was stopped with Ctrl+C to display the summary statistics.

```

64 bytes from 192.168.139.129: icmp_seq=26 ttl=64 time=0.745 ms
64 bytes from 192.168.139.129: icmp_seq=27 ttl=64 time=0.731 ms
64 bytes from 192.168.139.129: icmp_seq=28 ttl=64 time=2.10 ms
64 bytes from 192.168.139.129: icmp_seq=29 ttl=64 time=0.594 ms
64 bytes from 192.168.139.129: icmp_seq=30 ttl=64 time=1.06 ms
64 bytes from 192.168.139.129: icmp_seq=31 ttl=64 time=1.01 ms
64 bytes from 192.168.139.129: icmp_seq=32 ttl=64 time=1.13 ms
64 bytes from 192.168.139.129: icmp_seq=33 ttl=64 time=0.394 ms
64 bytes from 192.168.139.129: icmp_seq=34 ttl=64 time=1.15 ms
64 bytes from 192.168.139.129: icmp_seq=35 ttl=64 time=0.905 ms
64 bytes from 192.168.139.129: icmp_seq=36 ttl=64 time=0.921 ms
64 bytes from 192.168.139.129: icmp_seq=37 ttl=64 time=0.701 ms
64 bytes from 192.168.139.129: icmp_seq=38 ttl=64 time=0.769 ms
64 bytes from 192.168.139.129: icmp_seq=39 ttl=64 time=0.862 ms
64 bytes from 192.168.139.129: icmp_seq=40 ttl=64 time=0.731 ms
64 bytes from 192.168.139.129: icmp_seq=41 ttl=64 time=1.17 ms
64 bytes from 192.168.139.129: icmp_seq=42 ttl=64 time=1.31 ms
^C
— 192.168.139.129 ping statistics —
42 packets transmitted, 42 received, 0% packet loss, time 41183ms
rtt min/avg/max/mdev = 0.394/0.905/3.177/0.457 ms

```

Figure 3 — Successful ping from Kali to Metasploitable: 42 packets transmitted, 42 received, 0% packet loss

3.6 TryHackMe — DVWA Room

The free TryHackMe "DVWA" room was launched using the browser-based AttackBox. After extending the lab session and connecting to the assigned Lab Machine IP address, the DVWA login page was reached and accessed using the default DVWA credentials (admin / password). The room was completed at 100%.

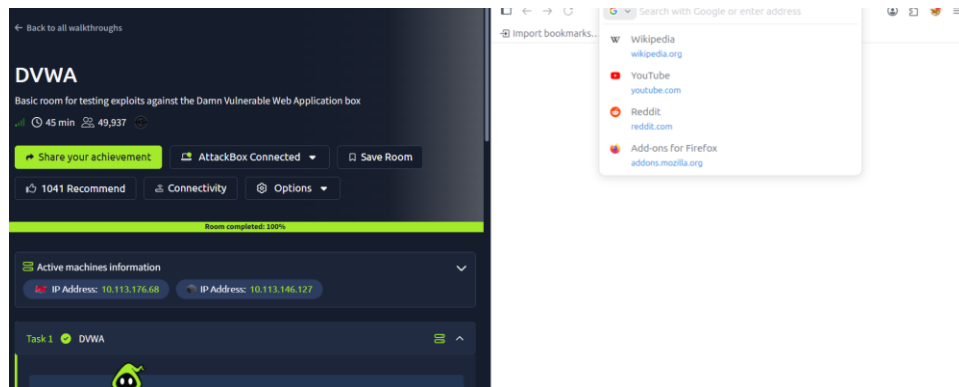


Figure 4 — TryHackMe DVWA room showing "Room completed: 100%"

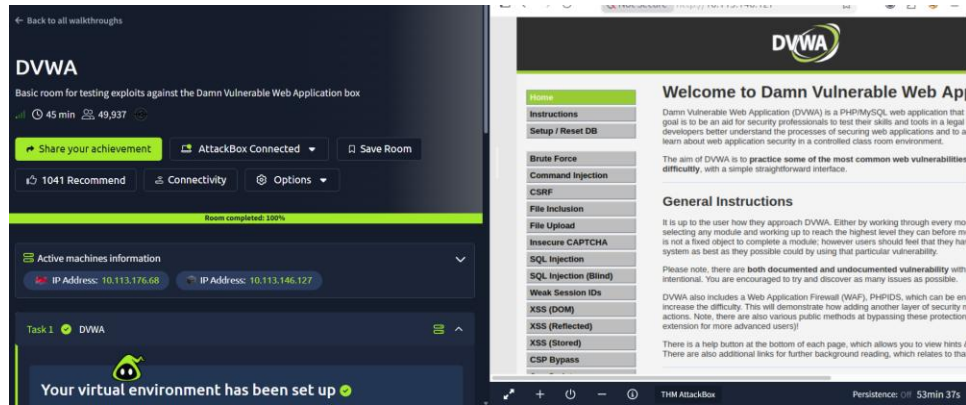


Figure 5 — DVWA dashboard successfully accessed via the TryHackMe AttackBox

4. Results Summary

Metasploitable 2 IP	192.168.139.129
Kali Linux IP	192.168.139.128
Ping Result	42 packets transmitted, 42 received, 0% packet loss
TryHackMe DVWA Room	Completed — 100%

5. Observations & Conclusion

- For two virtual machines to communicate, their network adapters must be configured to the same network mode (Host-only or NAT) so they share a common subnet.
- A successful ICMP ping (0% packet loss) confirms basic Layer-3 connectivity between the attacker and target machines, which is a required precondition before any scanning or exploitation activity in later tasks.
- Metasploitable 2 and DVWA are intentionally vulnerable systems and must only ever be run on an isolated, internal network — never exposed to an untrusted network or the public internet.
- The target IP addresses noted here (Metasploitable: 192.168.139.129) will be used as the scan target in upcoming Nmap/scanning tasks.

This completes Day 1 (practical setup) and Day 2 (reporting) of Week 2, Task 6.